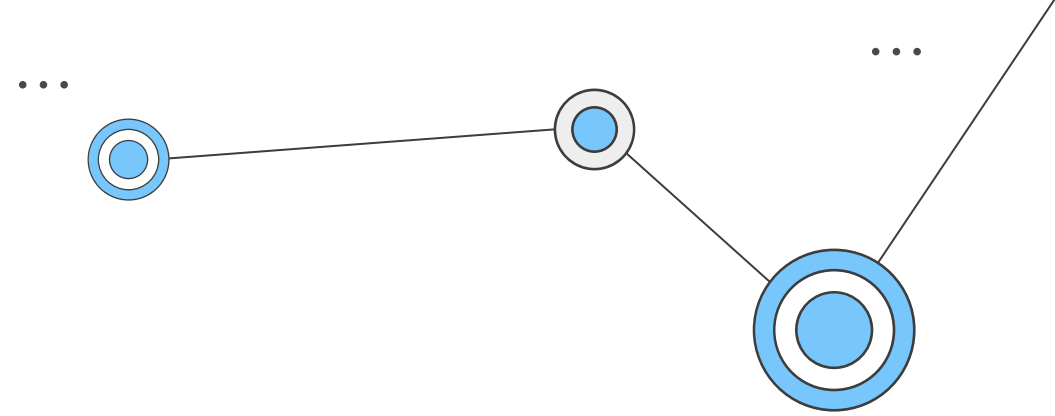
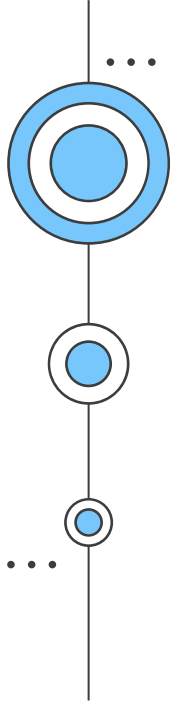




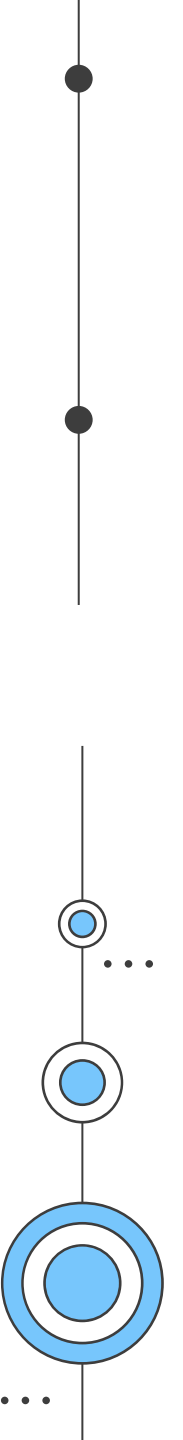
Ayudantía 01: Intro a Cloud

Martín Eyzaguirre

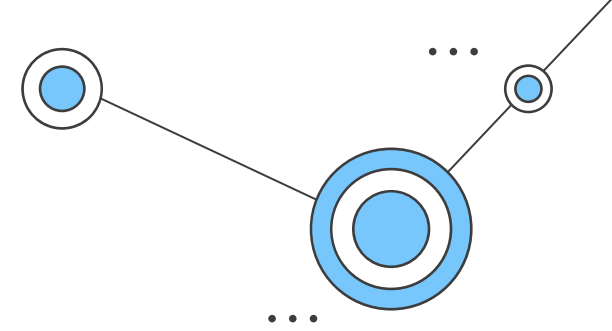


01

Cloud Computing



¿Por qué la nube?

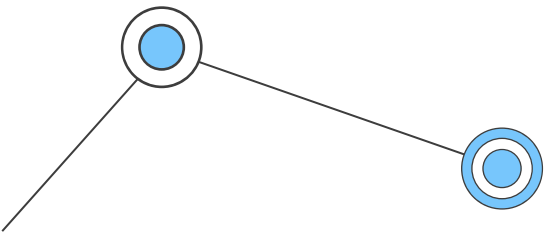


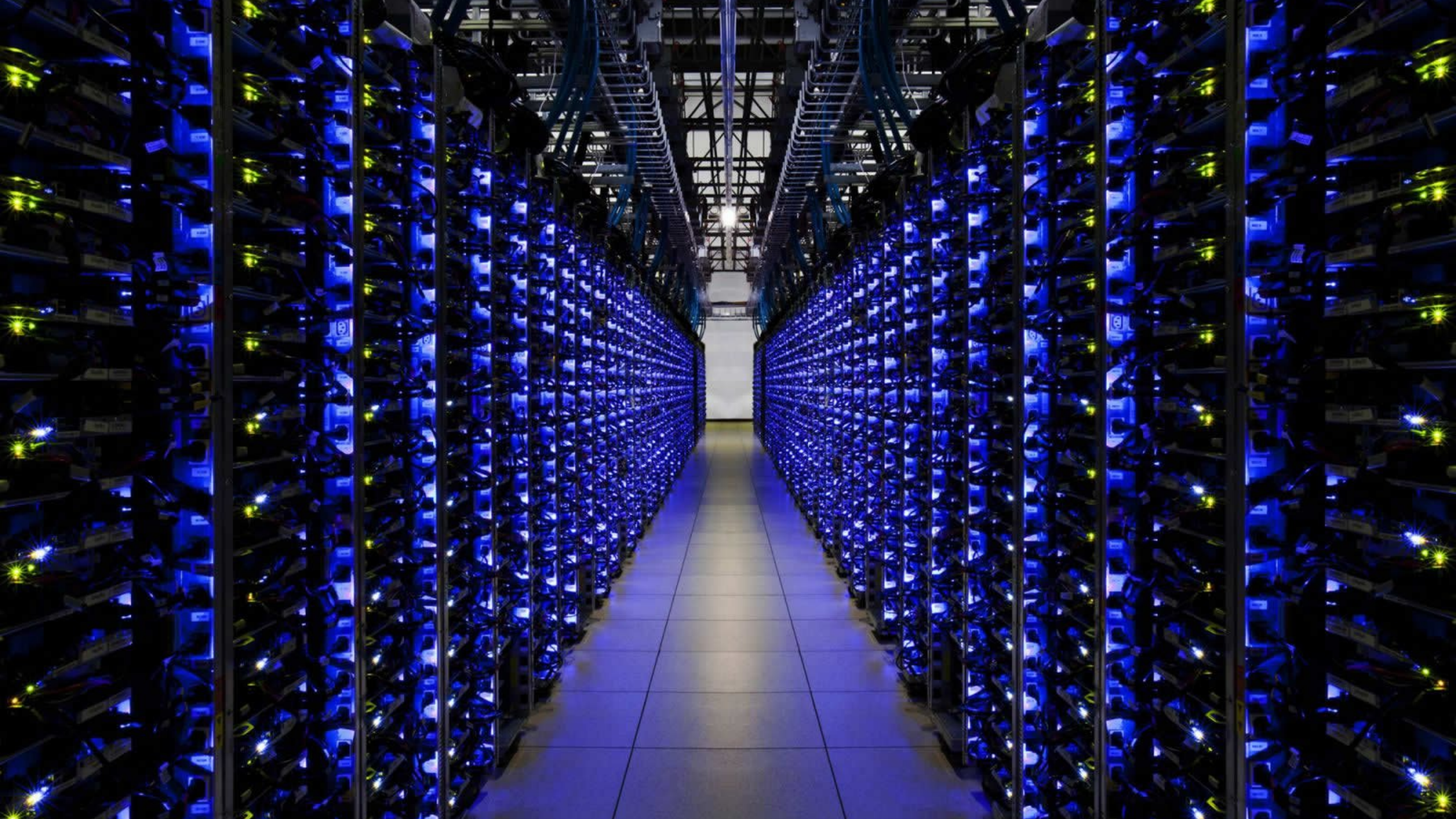
¿Tienen servidores personales?

¿Qué problemas aparecen?

¿Qué pasa si quiero servir a más clientes?

¿Cómo escalar en cantidad?





Pros

Servidores con 99.999% de disponibilidad

Seguridad de los servidores

Escalabilidad

Costos reducidos y adaptables

Infinidad de métodos de operar

Enorme oferta de servicios

Commodity

Virtualización

¿Quiénes ocupan esto?

Empresas
pequeñas

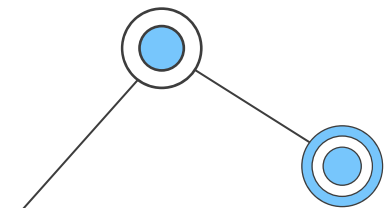
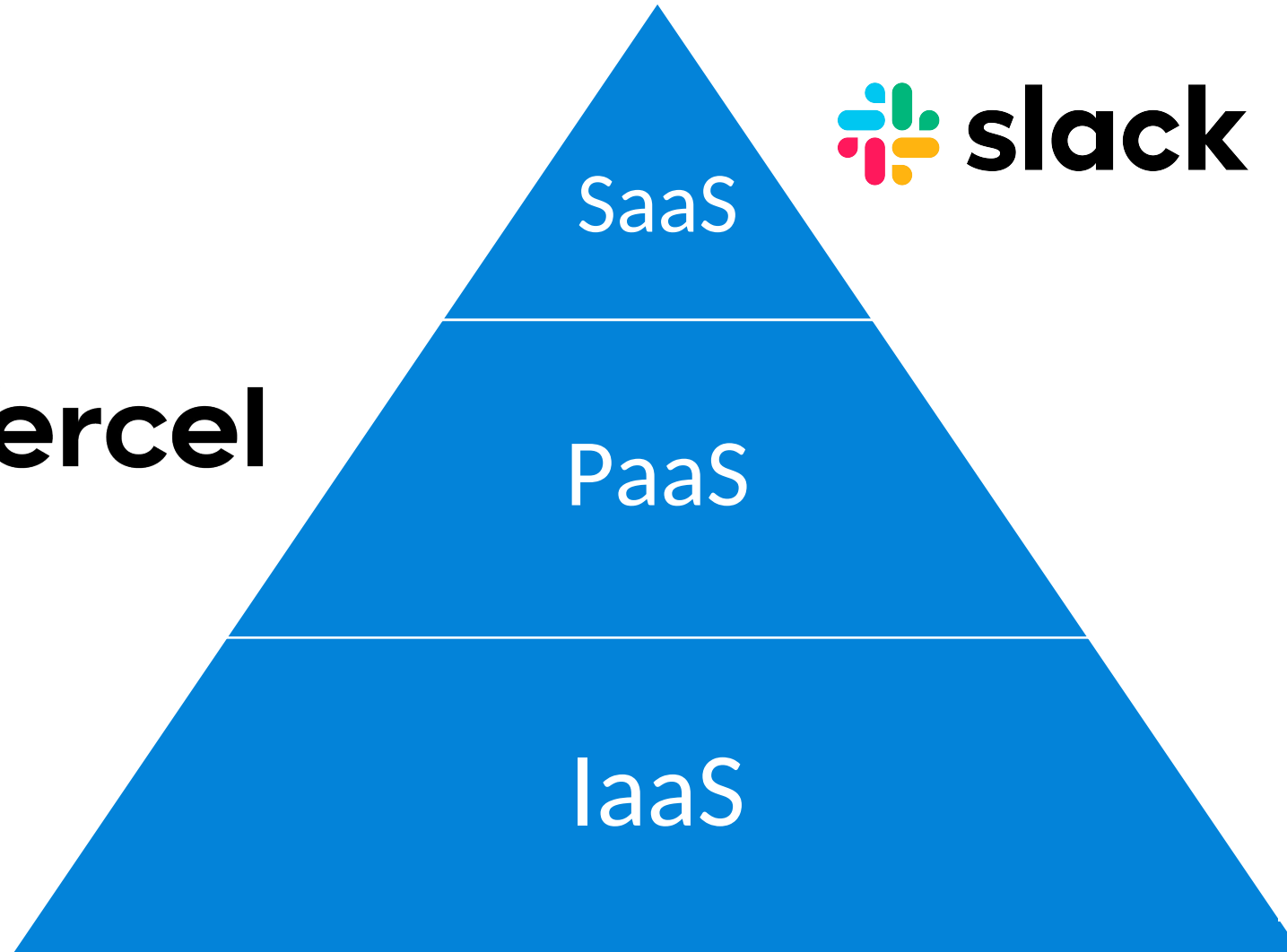
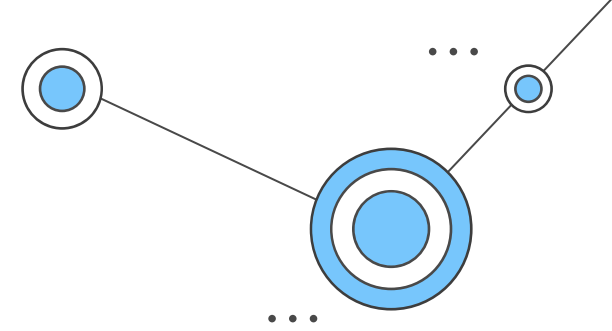
Empresas
medianas

Empresas
grandes

Empresas
mundiales

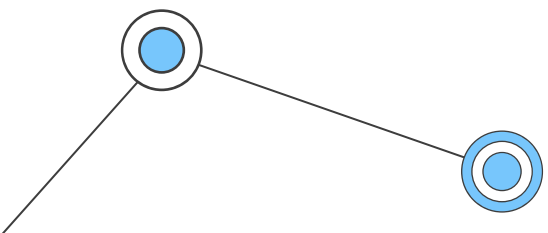
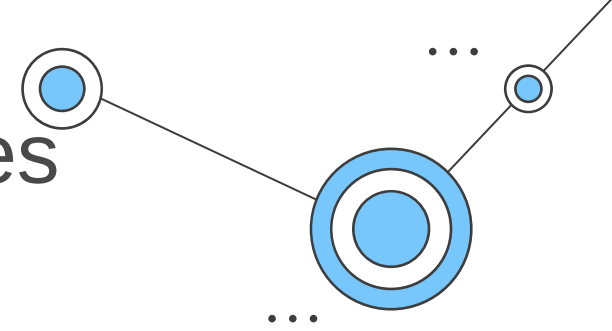
¡Personas!

Modelos de cloud computing



¿Qué nos permiten estos proveedores IaaS?

- Muchas opciones de infraestructura
- Almacenamiento
- Organizaciones de redes
- Servicios específicos
- Performance disponible
- Serverless
- Hosting
- Sistemas Operativos



Algunos proveedores

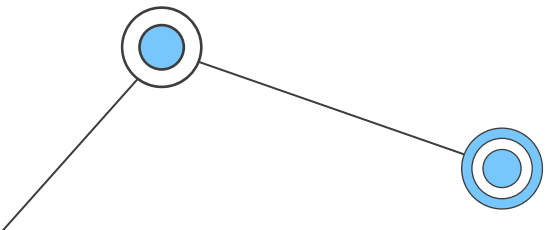
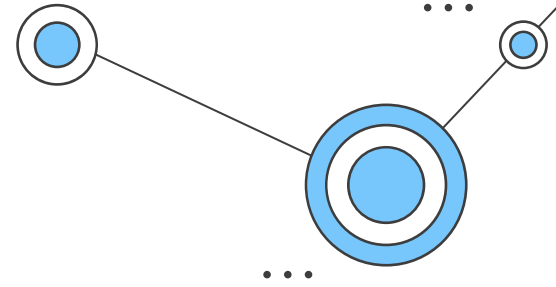


Google Cloud



¿En qué fijarse para operar en la nube?

- Flexibilidad
- Escalabilidad
- Performance
- Etc. que vieron en clases
- COSTO
- COSTO
- COSTO



¿Por qué usamos AWS?

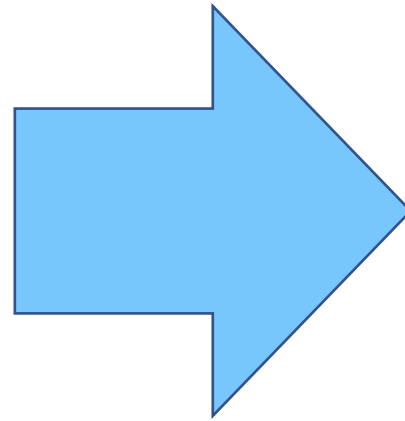


**Por
preferencia!**

AWS básico



Amazon EC2



ssh

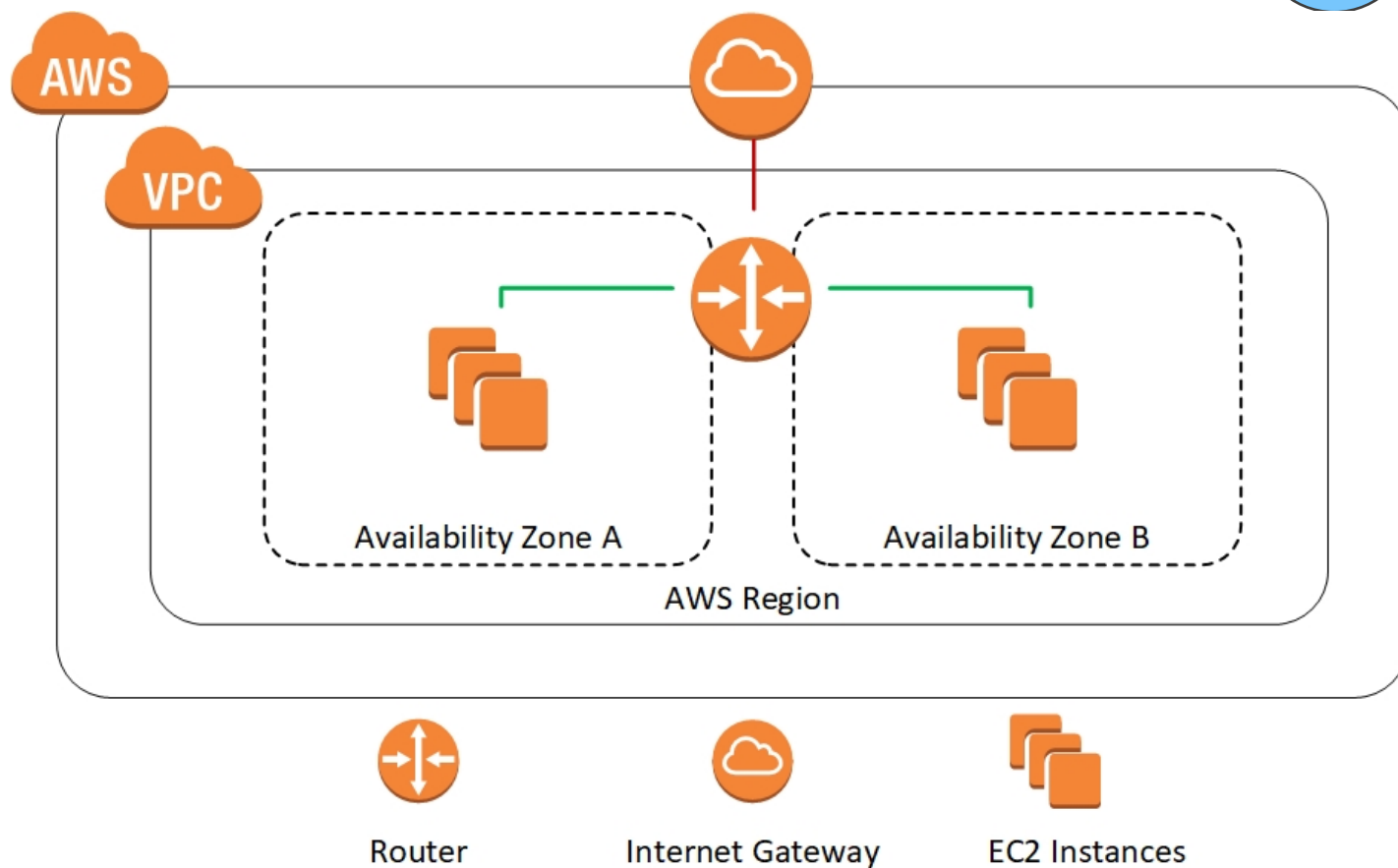
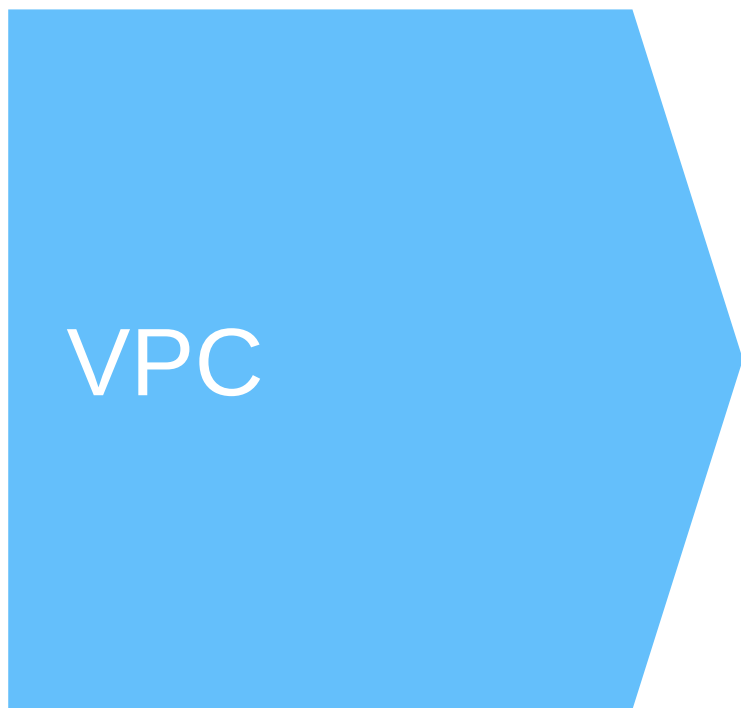


Tiers (t3)

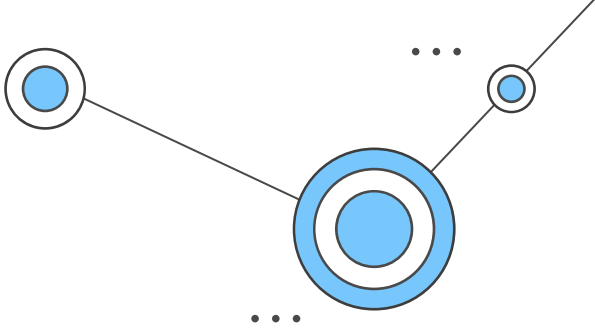
Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
t3.nano	\$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.micro	\$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.small	\$0.0208	2	2 GiB	EBS Only	Up to 5 Gigabit
t3.medium	\$0.0416	2	4 GiB	EBS Only	Up to 5 Gigabit
t3.large	\$0.0832	2	8 GiB	EBS Only	Up to 5 Gigabit
t3.xlarge	\$0.1664	4	16 GiB	EBS Only	Up to 5 Gigabit
t3.2xlarge	\$0.3328	8	32 GiB	EBS Only	Up to 5 Gigabit

Tiers (c6i)

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
c6i.large	\$0.085	2	4 GiB	EBS Only	Up to 12500 Megabit
c6i.xlarge	\$0.17	4	8 GiB	EBS Only	Up to 12500 Megabit
c6i.2xlarge	\$0.34	8	16 GiB	EBS Only	Up to 12500 Megabit
c6i.4xlarge	\$0.68	16	32 GiB	EBS Only	Up to 12500 Megabit
c6i.8xlarge	\$1.36	32	64 GiB	EBS Only	12500 Megabit
c6i.12xlarge	\$2.04	48	96 GiB	EBS Only	18750 Megabit
c6i.16xlarge	\$2.72	64	128 GiB	EBS Only	25000 Megabit
c6i.24xlarge	\$4.08	96	192 GiB	EBS Only	37500 Megabit
c6i.32xlarge	\$5.44	128	256 GiB	EBS Only	50000 Megabit
c6i.metal	\$5.44	128	256 GiB	EBS Only	50000 Megabit



Code	Name	AZs	Geography	Opt-in status
us-east-1	US East (N. Virginia)	6	United States of America	Not required
us-east-2	US East (Ohio)	3	United States of America	Not required
us-west-1	US West (N. California)	3 †	United States of America	Not required
us-west-2	US West (Oregon)	4	United States of America	Not required
af-south-1	Africa (Cape Town)	3	South Africa	Required
ap-east-1	Asia Pacific (Hong Kong)	3	Hong Kong	Required
ap-south-2	Asia Pacific (Hyderabad)	3	India	Required
ap-southeast-3	Asia Pacific (Jakarta)	3	Indonesia	Required
ap-southeast-5	Asia Pacific (Malaysia)	3	Malaysia	Required
ap-southeast-4	Asia Pacific (Melbourne)	3	Australia	Required
ap-south-1	Asia Pacific (Mumbai)	3	India	Not required
ap-southeast-6	Asia Pacific (New Zealand)	3	New Zealand	Required
ap-northeast-3	Asia Pacific (Osaka)	3	Japan	Not required
ap-northeast-2	Asia Pacific (Seoul)	4	South Korea	Not required
ap-southeast-1	Asia Pacific (Singapore)	3	Singapore	Not required
ap-southeast-2	Asia Pacific (Sydney)	3	Australia	Not required
ap-east-2	Asia Pacific (Taipei)	3	Taiwan	Required
ap-southeast-7	Asia Pacific (Thailand)	3	Thailand	Required
ap-northeast-1	Asia Pacific (Tokyo)	4	Japan	Not required
ca-central-1	Canada (Central)	3	Canada	Not required
ca-west-1	Canada West (Calgary)	3	Canada	Required
eu-central-1	Europe (Frankfurt)	3	Germany	Not required



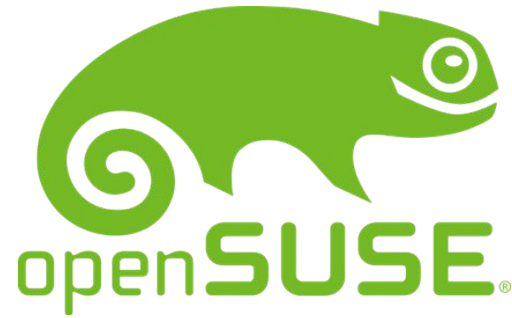
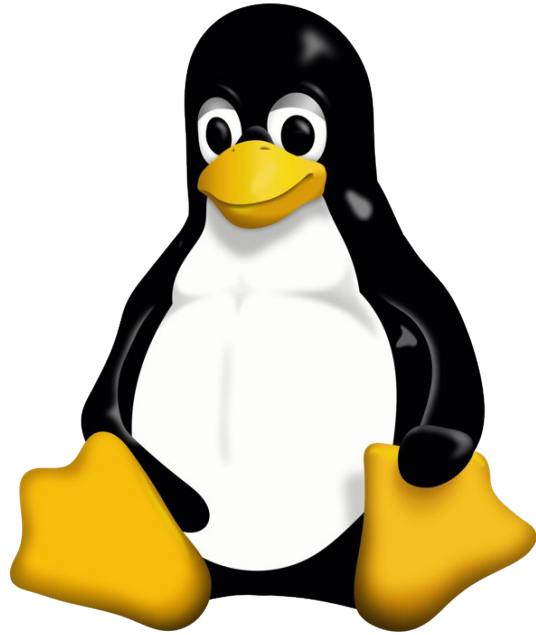
eu-south-1	Europe (Milan)	3	Italy	Required
eu-west-3	Europe (Paris)	3	France	Not required
eu-south-2	Europe (Spain)	3	Spain	Required
eu-north-1	Europe (Stockholm)	3	Sweden	Not required
eu-central-2	Europe (Zurich)	3	Switzerland	Required
il-central-1	Israel (Tel Aviv)	3	Israel	Required
mx-central-1	Mexico (Central)	3	Mexico	Required
me-south-1	Middle East (Bahrain)	3	Bahrain	Required
me-central-1	Middle East (UAE)	3	United Arab Emirates	Required
sa-east-1	South America (São Paulo)	3	Brazil	Not required

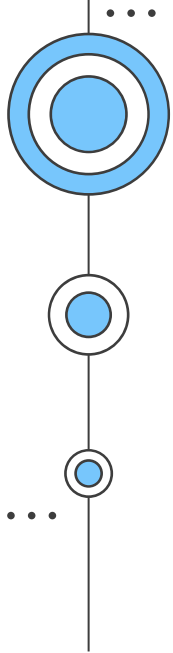
Sistemas Windows disponibles



Windows Server

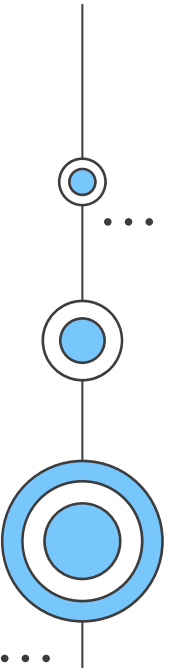
Sistemas Linux disponibles





02

Linux



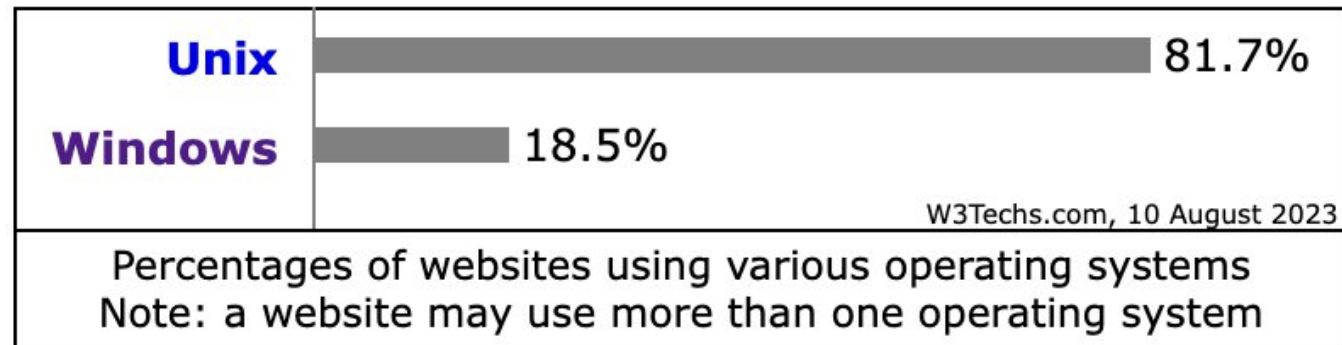
Linux

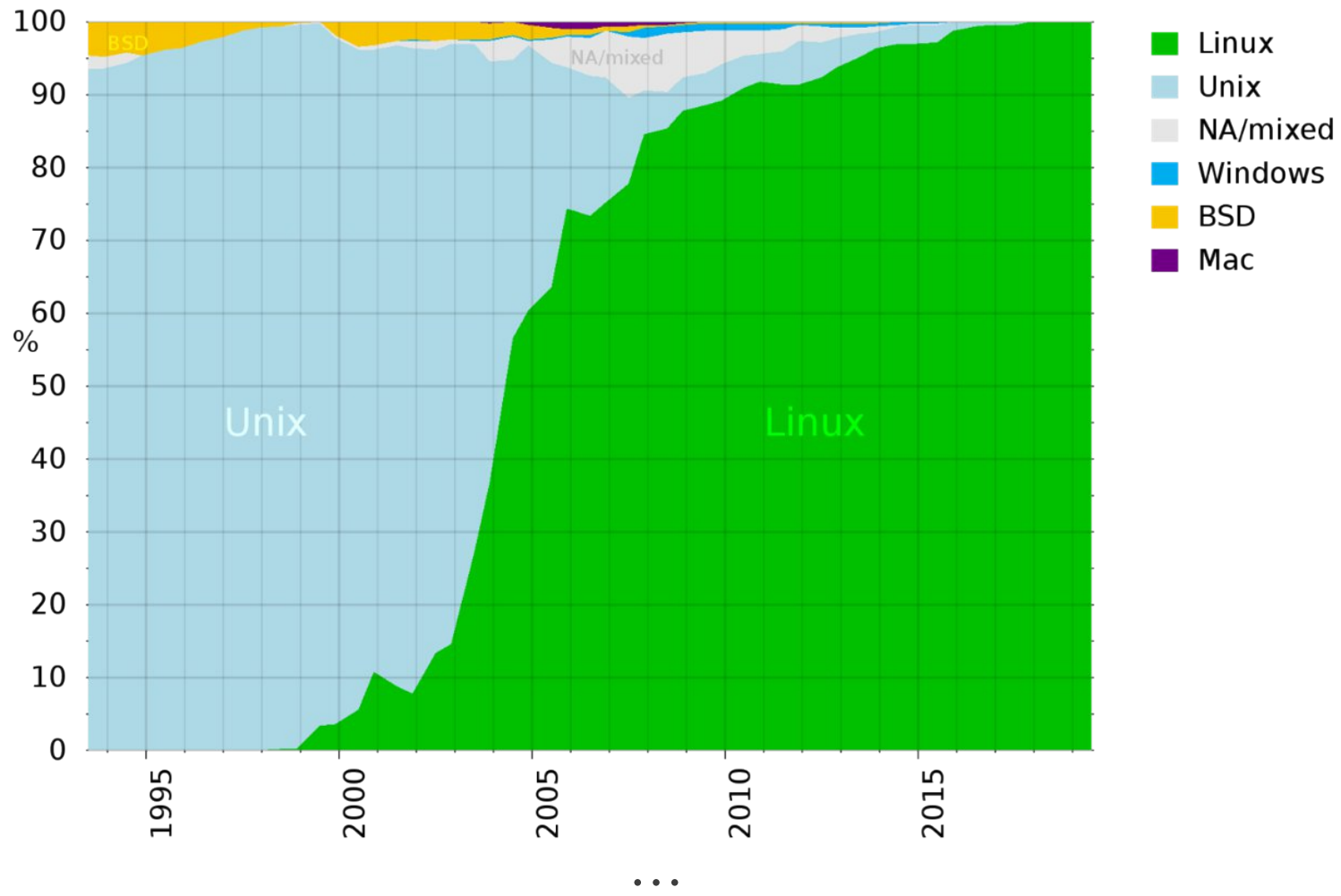
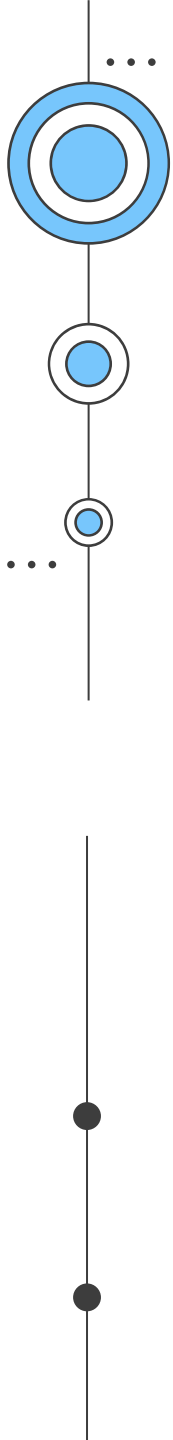


- Omnipresente en el mundo de los servidores
- Estable, seguro, versátil, eficiente
- Documentaciones y tutoriales disponibles
- Ubuntu server -> Abordable

How to read the diagram:

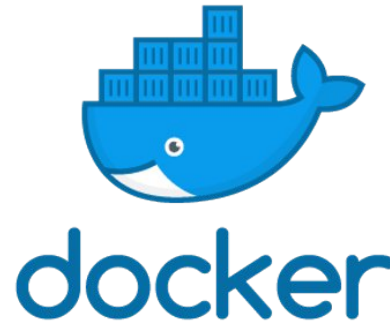
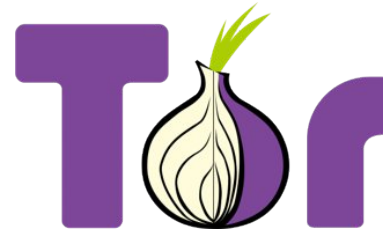
Unix is used by 81.7% of all the websites whose operating system we know.



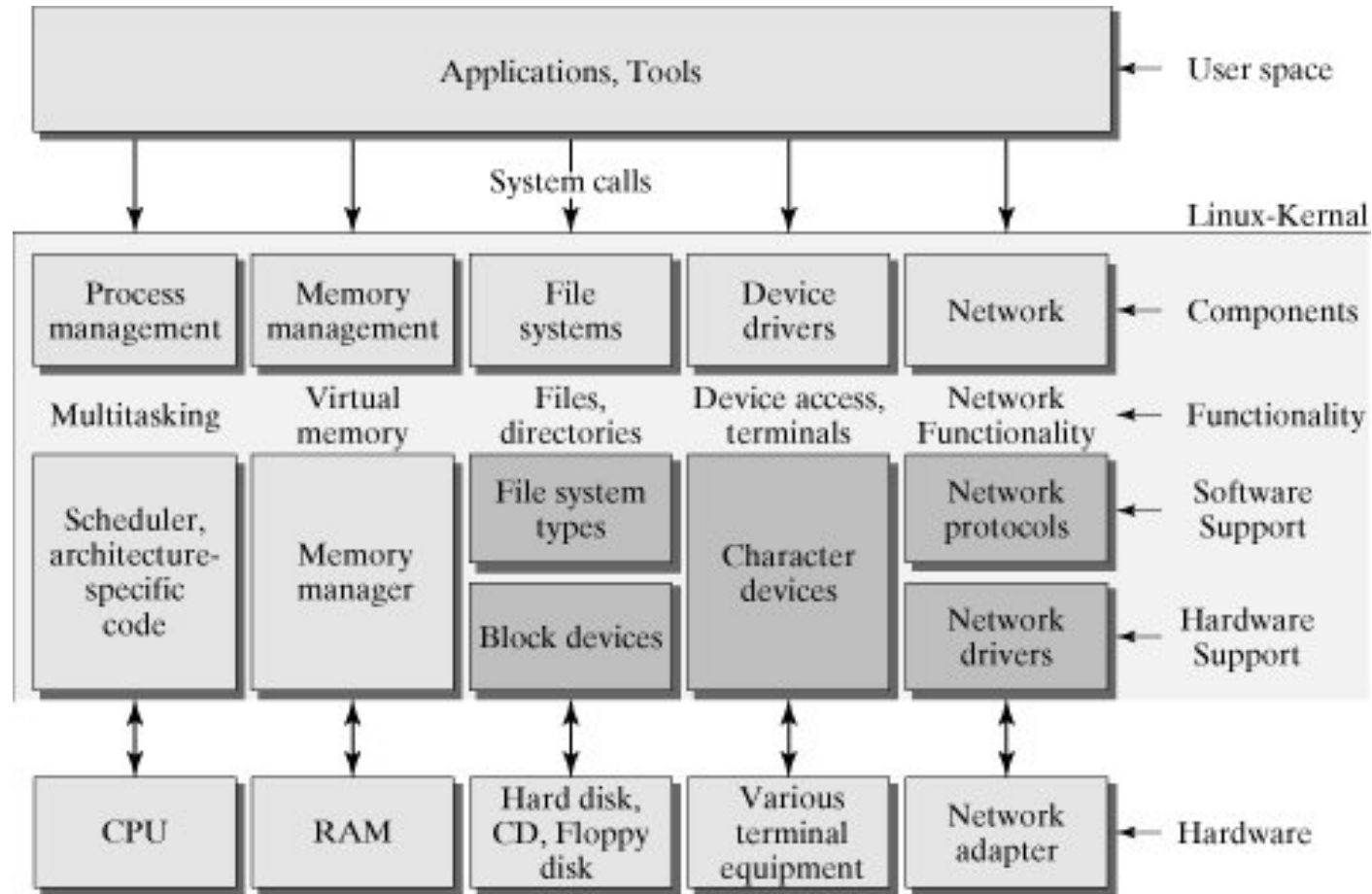


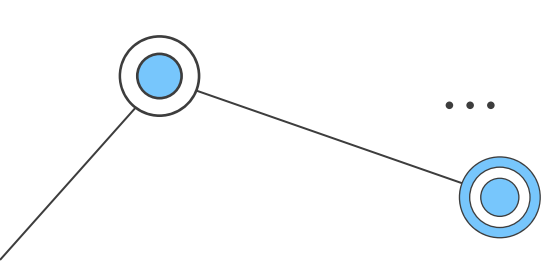
Comandos y herramientas de Linux interesantes

- Cron
- Bash
- Man
- ssh
- Wget
- Sudo
- Grep
- Curl



Arquitectura de Linux





```
ubuntu@ip-172-31-62-120: ~$
```



Bash

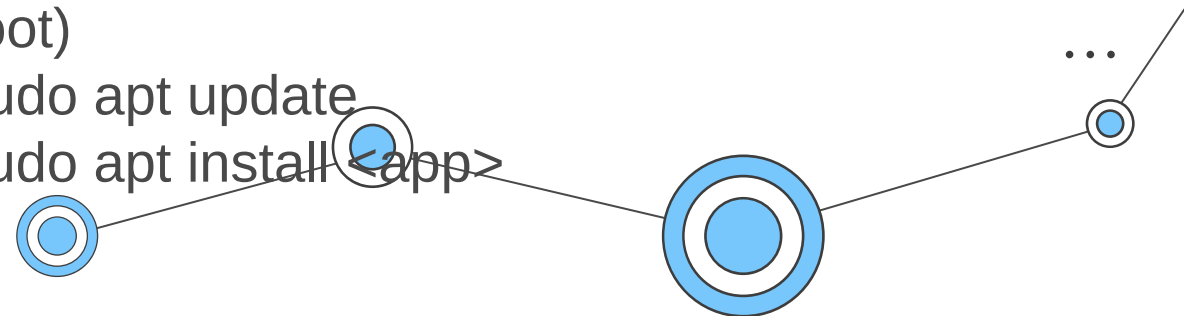
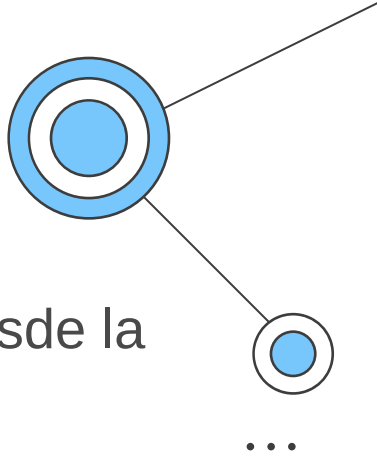
Shell para interactuar con el usuario desde la línea de comandos

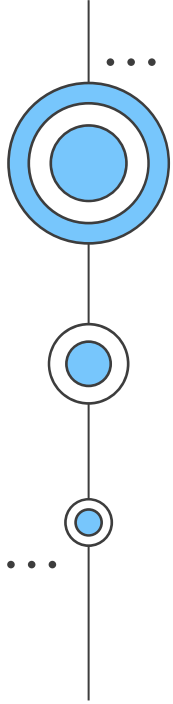
Estándar de facto

Alternativas: zh, sh, csh (les aconsejamos Bash)

Comandos útiles:

- `cd <dir>` (Cambia de directorio)
- `mkdir <name>` (Crea un directorio)
- `sudo <command>` (Ejecuta comando como root)
- `sudo apt update`
- `sudo apt install <app>`

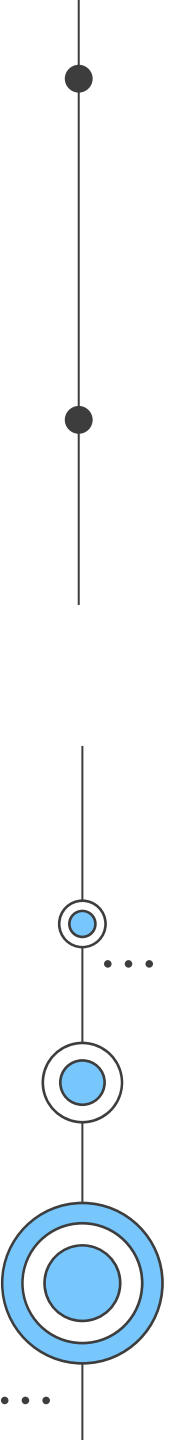




03

Dentro del servidor

r



¿Cómo nos conectamos al servidor?: Por SSH

La forma estándar de acceder a servidores remotos

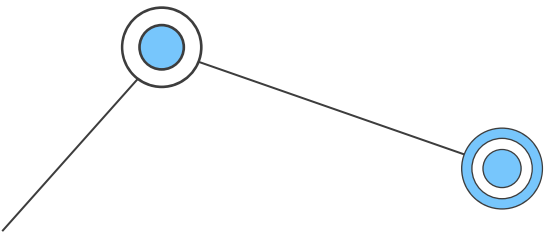
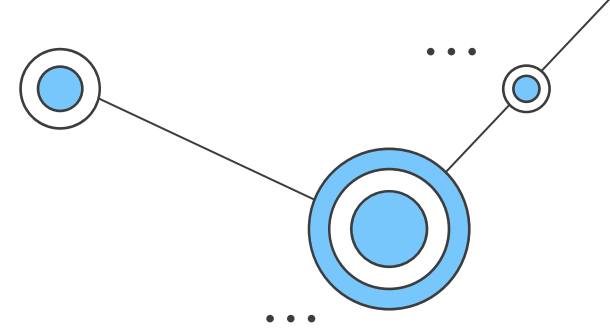
En windows pueden usar: Putty

En Linux es nativo (no requiere instalación)

Crea una conexión segura entre computadores

```
ssh -i <key> <user>@<ip>
```

En sus servidores usarán un archivo como llave (debe tener permiso 400)



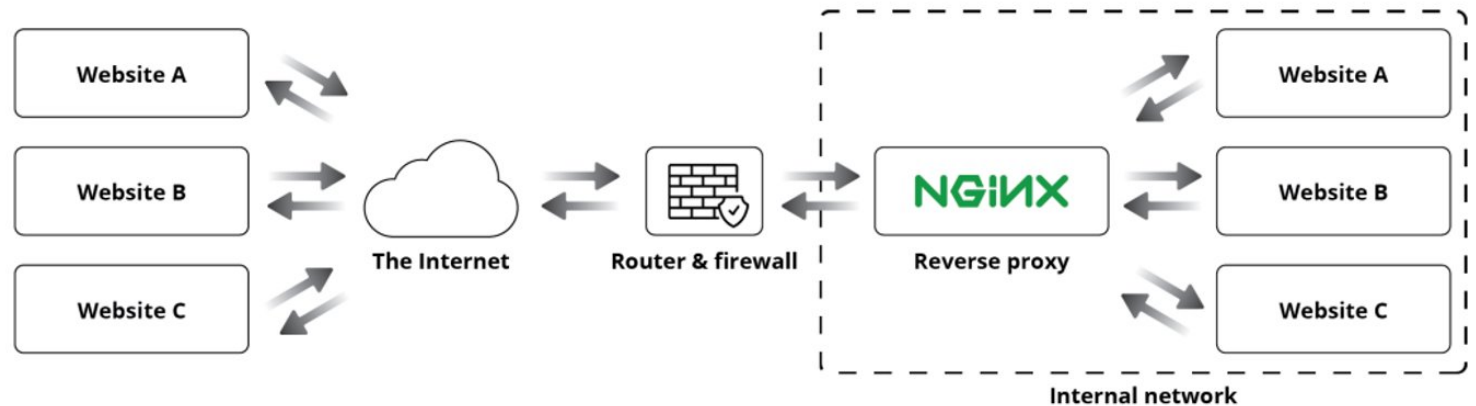
¿Cómo habilitamos un servidor para disponibilizar nuestras apps?

Tiene varias funcionalidades

- Web server
- Load Balancer
- **Reverse Proxy**
- E incluso más

Pueden instalarlo fácilmente, pero necesita configuración dentro del servidor

NGINX



Otros paradigmas

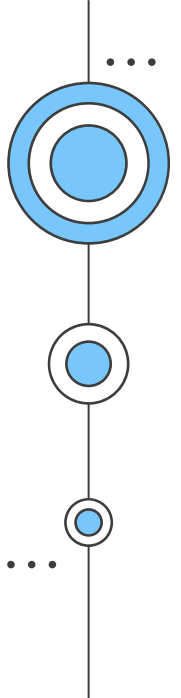


Amazon
Lambda

- Costo por ejecución y RAM
- Sumamente eficiente

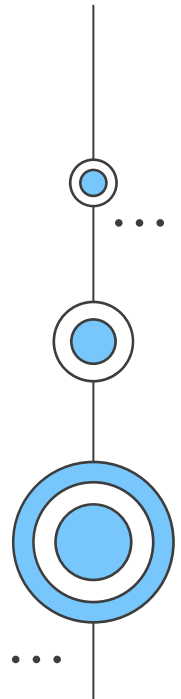


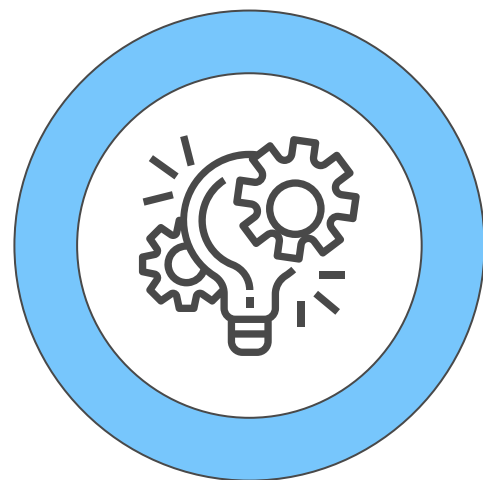
- Muy fácil de escalar
- "Scaffolding" de infraestructura en AWS (PaaS)
- No lo podrán usar en su proyecto 😞



04

¡Vamos a
la consola!





Dudas

...